



travels through the magnetic field. Current is collected by electrodes, which are in contact with the plasma jet. Electrical properties of the plasma are optimised by seeding the explosive or propellant with suitable additives.

Low-frequency e-bombs configured around FCGs or MHD generators are far less lethal than HPM e-bombs, as a large cross-section of targets are difficult to attack even with very high power levels at their output frequencies. Focussing the output energy from these devices is also a huge problem.

On the other hand, microwave weapons not only couple more readily to a variety of targets than low-frequency weapons, their output can be tightly focussed. Though there are a host of microwave sources including relativistic klystrons, travelling wave tubes, magnetrons, spark-gap devices and vircators, from the perspective of a microwave e-bomb's designer, vircator by virtue of being a one-shot device capable of producing an extremely powerful single pulse of radiation is the preferred choice as of now. In addition, it is mechanically simple, small and robust. Spark-gap devices may be used in the near future. Fig. 7 shows the component parts of an HPM e-bomb.

Lethality of e-bombs

Unlike conventional kinetic energy weapons, lethality analysis in case of HPM weapons is a far more complex proposition. There are two principal coupling modes by which electromagnetic energy released by the weapon is coupled into the target. These are front-door coupling and back-door coupling. In the case of the former, energy is coupled to the targeted equipment such as radar or communications equipment through the antenna. Back-door coupling occurs through

large transient currents produced in the case of low-frequency weapons or electrical standing waves on fixed electrical wiring and cables interconnecting equipment, or providing connections to mains power in the case of HPM weapons.

Representative HPM weapon systems

Based on the state-of-the-art in HPM weapons of various categories and emerging trends, HPM devices will be available for deployment in de-

radars, satellites and command, control and communications infrastructure. Some well-known developed and under-development HPM systems are Active Denial System (ADS) and Vigilant Eagle of Raytheon, HPM Blackout built by BAE Systems, RANETS-E of Russian origin, Shortstop Electronic Protection System (SEPS) of Whittaker Corp., Warlock Green and Warlock Red of EDO Communications and Countermeasures, DS-110 of Diehl, Germany, and CHAMP developed jointly by US Air Force Research Laboratory (AFRL) and Boeing Phantom Works.

ADS. It is a non-lethal microwave-directed energy system that can be used in counter-personnel role against hostile human targets at distances beyond the effective range of small arms. Active denial technology is a breakthrough non-lethal technology that uses millimetre-wave electromagnetic energy for area denial, perimeter security and riot-control applications from relatively longer ranges without application of any lethal force. With a range of about 500 metres and beam that is a little less than two metres across, ADS would be capable of dispersing riots from a safe distance.

It has been developed by Raytheon for the US Air Force Research Labs and Department of Defence Joint Non-lethal Weapons Directorate. It operates as follows.

Microwave energy at 95GHz generated by a gyrotron is focused and directed at the targeted personnel by a directional planar antenna. The microwave energy produces an intolerable heating sensation within seconds, forcing the subject to flee from the scene to avoid exposure to the microwave energy beam. The millimetre-wave energy, in fact, heats up water molecules just under the subject's skin, heating these enough to cause extraordinary pain.



Fig. 9: Silent Guardian protection system

fensive and offensive roles by 2030s. These weapons would be mounted on fixed and mobile land and ship based platforms as part of an integrated air or point defence. There will also be airborne defensive systems, including self-protection suites to target enemy radars and missiles and HPM warheads on air-to-air missiles to target attacking enemy aircraft and missiles.

Offensive HPM weapons will include precision-guided e-bombs, cruise missiles, unmanned or manned aircraft with integrated HPM weapons. These offensive weapons will be used to target